

John P. Acosta

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EDUCATION

NEW JERSEY INSTITUTE OF TECHNOLOGY

Newark, NJ

Bachelor of Science in Computer Science, Ying Wu College of Computing

Expected May 2027

- **GPA:** 3.76 **Awards:** NSF S-STEM Scholar, Dean's List (All Semesters), Ron Andrews PGIM Scholar, Kupfrian Memorial Scholar
- **Relevant Coursework:** Data Structures & Algorithms, Object-Oriented Programming, Programming Language Concepts (C/C++), Operating Systems, Computer Networks, Artificial Intelligence, Data Science, Database Design & Management, Linear Algebra, Probability & Statistics

SKILLS

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| Languages: Python, Java, C/C++, SQL | Data/ML: NumPy, pandas, scikit-learn, PyTorch, Transformers, PEFT |
| Frameworks: FastAPI, React/Next.js | Cloud/DevOps: AWS, Azure, Terraform, Jenkins, GitHub Actions |
| Databases: PostgreSQL, SQLite, Supabase | Tools: Git/GitHub, Linux/Bash, Docker, LangChain, SLURM/HPC |

PROFESSIONAL EXPERIENCE

Software Engineer Intern

Summer 2026

PGIM Technology DevOps

- Currently migrating infrastructure templates from AWS CloudFormation to Terraform, enabling cross-platform deployment across AWS and Azure across Prudential Technology and reducing per-build scan overhead from 30 security scans to 1, targeting significant reduction in build time

Applied AI Research Engineer Intern

Jan 2026 - Present

NJIT AIM Lab

- Designed a QA framework across a PyQt desktop application, authoring 100 natural-language prompt suites to capture UI screenshots and telemetry on shared HPC infrastructure, identifying a 20% pass rate that drove improvements to model accuracy and prompt handling
- Prototyped LangChain agentic workflows connecting segmentation models to a surgeon-facing interface, validating feasibility for clinical deployment
- Delivered weekly technical presentations to biomedical experts, conducting EDA across BraTS medical imaging datasets to inform model selection and architecture recommendations for an agentic tumor segmentation system

Undergraduate Research Assistant

Jan 2026 - Present

Lijing Wang Lab | NJIT

- Co-authoring a journal paper on training instability and variance in LLM fine-tuning pipelines, designing controlled isolation experiments across Llama 3.2-3B, RoBERTa, and BERT on NJIT's Wulver A100 HPC cluster
- Diagnosed and resolved multiple sources of RNG inconsistency across data ordering, model initialization, and dropout in the training pipeline, preventing invalidation of 400+ planned runs by redesigning affected components with per-factor seed control

Deep Learning Forecasting Research Intern

Aug 2025 - Jan 2026

Grace Hopper AI Research @ NJIT

- Engineered modular Python preprocessing routines (NumPy, pandas) for cleaning, trimming, and interpolating 10+ multi-location epidemic datasets (~900 weekly observations each), creating consistent train/test splits and enabling reliable rolling-origin forecasting across 5 disease types
- Parallelized forecasting workflows with multi-threaded Python execution, reducing runtime by ~80% and enabling rapid model experimentation across diseases and hyperparameters without HPC resources
- Built project-wide logging utilities to automate metric collection and mid-run recovery, replacing manual aggregation and enabling reproducible experimentation at scale across datasets and model families

TECHNICAL PROJECTS

SHPE NJIT APP | Software Developer, CI/CD | React Native, Supabase, GitHub Actions

- Own CI/CD for the SHPE NJIT chapter mobile app, architecting automated test and deploy pipelines via GitHub Actions to support 200+ active members

AI-TUTOR | Lead Developer | Python, GraphRAG, Docker, Flask, BeautifulSoup

- Developed a pedagogical tutor mapping concept prerequisites via a knowledge graph built from a custom BFS web crawler and nano-graphrag extended with domain-specific entity extraction, grounding responses in graph-retrieved context to prevent hallucinations
- Implemented a rate-limited LLM wrapper with exponential backoff for reliable request handling, exposed via Flask backend and Streamlit UI on a Docker devcontainer configured for selective service loading

EXTRACURRICULAR ACTIVITIES

HISPA (Hispanics Inspiring Students' Performance and Achievements) | Member

Sep 2025 – Present

Break Through Tech AI Fellow | Cornell Tech

Jun 2025 – Present

Society of Hispanic Professional Engineers | NJIT

Jan 2024 – Present

Google Software Engineering Fellow (Selected 2x) | G-SWEP

Jul 2025 – Sept 2025